



BATHSHEVA LONDON



Faro

Prototype build manual

INTRODUCTION

Before you begin

From the box of printed parts to a finished, glowing prototype ready to photograph.

Faro is built from seventeen printed pieces, finished by hand in four colours and glued together in a set order. Nothing here is difficult, but each stage needs patience: thin coats of paint, time to dry, and a dry fit before any glue. Each step ends with a checkpoint; don't move on until it is met.

TIMING

Weekend one

Inspect and dry-fit, prepare the surfaces, prime, and paint the colours.

The week between

Let the paint harden. It feels dry within hours but needs days to cure.

Weekend two

Clear coat, assemble, light it and photograph.

SAFETY

- Spray outdoors or somewhere very well ventilated, and wear the respirator whenever you spray.
- Wear a dust mask or the respirator, and nitrile gloves, when sanding resin: its dust shouldn't be breathed in or left on skin.
- Wear nitrile gloves when mixing or applying epoxy.
- Keep sprays and epoxy away from children and pets, and let fumes clear before bringing parts indoors.

HOW TO USE THIS MANUAL

Work through the seven steps in order. Each part's colour and treatment is listed on the next page; keep it open while you paint. If something goes wrong, turn to Troubleshooting at the back: almost every mistake can be sanded out and repainted.



INVENTORY

Parts and supplies

PART	QTY	FINISH
 base	1	Black gloss, clear coat
 base_plate	1	Black acrylic, or leave primed
 nameplate	1	Gold, black round the letters
 band_cream	1	Cream gloss, clear coat
 band_red	1	Red gloss, clear coat
 tower	1	Cream gloss, clear coat
 knob	1	Gold; decorative, it doesn't turn
 gallery	1	Gold, light coats only
 lantern_frame	1	Gold, light coats only
 cap	1	Red gloss, clear coat
 finial	1	Gold
 lantern_glass	1	Leave unpainted
 window_diffuser_1 to 5	5	Leave unpainted

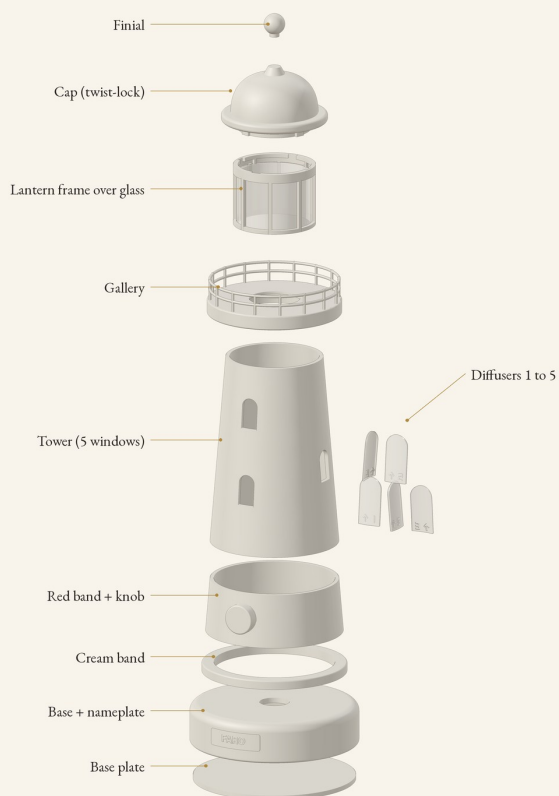
SUPPLIES

- Grey filler primer
- Red gloss spray
- Cream gloss spray (Ivory Bisque)
- Metallic gold spray
- Gloss black spray
- Clear lacquer
- Black acrylic and a small brush
- Wet-and-dry paper, P240 to P1200
- Tamiya masking tape, 10 mm
- Araldite Rapid epoxy, cocktail sticks
- Black self-adhesive felt
- Warm white copper fairy lights, 3 AA batteries
- Rechargeable warm white LED puck, under 45 mm
- A few coins for weight
- Respirator, dust mask, nitrile gloves
- Craft knife, blu-tack

STEP 01

Inspect and dry-fit

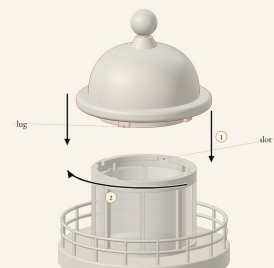
Check every part and fit them together without glue. Paint adds thickness, so tight fits only get tighter.



- 01 Tick off every part against the inventory. Look for cracks, warping or broken railing posts.
- 02 Stack without glue: base, cream band, red band, tower, gallery. Each joint should sit flat and centred.
- 03 Stand the lantern glass on the gallery and lower the frame over it. It should sit without forcing.
- 04 Twist the cap on. Under it are four lugs, small tabs that drop through four slots in the lantern top; then turn it clockwise to the stop.
- 05 Hold each diffuser behind its window: its number and arrow face inwards, arrow up. They are not interchangeable.
- 06 Press the base plate into the rebate: the shallow step cut round the opening under the base.
- 07 Stand it where it would live and judge size and proportions.



Dry-stacked: knob and nameplate aligned on the front.



Lugs over the slots: drop the cap in (1), turn clockwise to the stop (2).

READY TO MOVE ON WHEN

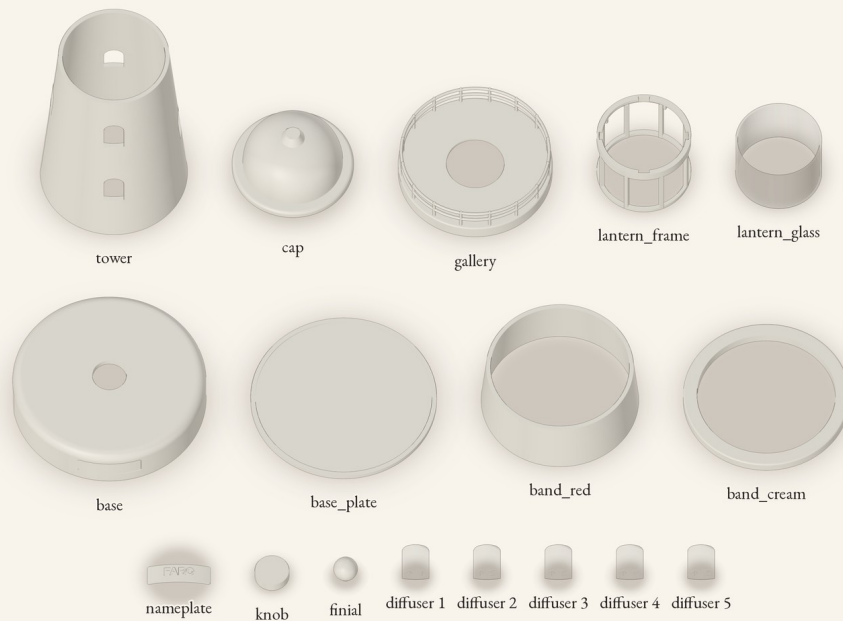
every part is present and undamaged, the stack stands straight, and the cap twists on and off without forcing.

STEP 02

Prepare the surfaces

Clean, smooth surfaces are what make the paint look expensive.

- 01 Wash all parts in warm soapy water, rinse, and leave to dry fully.
- 02 Trim any support nubs with a craft knife, cutting away from yourself.
- 03 Put on a dust mask or the respirator, and gloves. Sand visible surfaces with P240 to flatten marks, then P400 to smooth.
- 04 Wipe the dust off with a barely damp cloth and let dry.



Every printed part, as it arrives in plain white resin.

HANDLE WITH CARE

- Gallery and lantern frame: only remove support marks; the posts are delicate.
- Nameplate: sand the flat face only, never across the raised letters.
- Lantern glass and diffusers: never sand; they arrive frosted. Keep the small raised number and arrow on each diffuser. Set them aside, clean.

READY TO MOVE ON WHEN

every part is clean, dry and dust-free, with no support marks left on the surfaces that show.

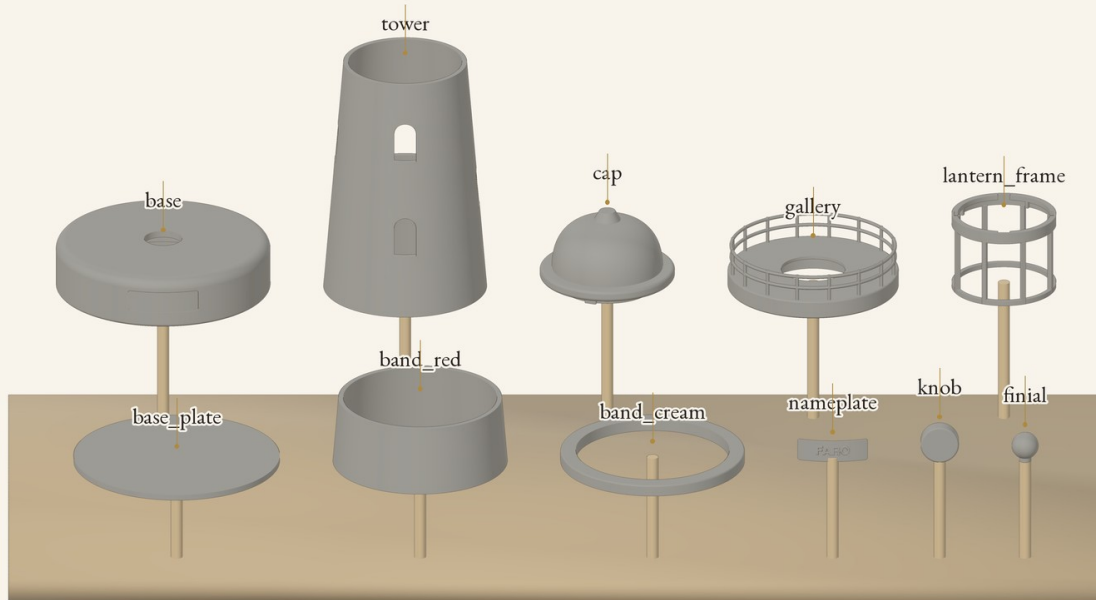
STEP 03

Prime

Primer fills fine layer lines and shows every flaw. This is where the final finish is really made.

- 01 Mount parts on a tube, bottle or cocktail sticks with blu-tack, so you can reach all round without touching.
- 02 Choose a dry, still day, around 15 to 25°C. Shake the primer for two to three minutes.
- 03 Spray light coats from 25 to 30 cm, starting and finishing each pass off the part.
- 04 Let it dry, then sand with P400, wearing the dust mask. Repeat until the surfaces look smooth under a bright light.
- 05 Prime every part except the lantern glass and diffusers. On the gallery and frame, one or two mist coats only.

Drying times. Typically touch-dry in 15 to 30 minutes, ready for the next coat after about 30 minutes, and ready to sand after at least an hour. Brands vary, so check the can.



Every part mounted for priming, except the glass and diffusers.

READY TO MOVE ON WHEN

every part is an even grey, with no layer lines, pits or bare spots under a bright light.

STEP 04

Paint

*Every part is a single colour, so no masking between colours is needed.
Tape only the surfaces that will be glued.*

COLOUR	PARTS
Red gloss	band_red, cap
Cream gloss	band_cream, tower
Metallic gold	gallery, lantern_frame, knob, finial, nameplate
Black gloss	base
Black	base_plate, and the recesses round the letters



MASK BEFORE PAINTING

- The band rims: the top and bottom faces of the cream and red bands
- The tower's bottom rim
- The gallery's underside
- The back of the knob
- The nameplate recess on the base
- The lantern frame's bottom ring

Glue holds to bare plastic far better than to paint. Peel the tape once the last coat is touch-dry.

SPRAYING

- 01 Three or four light coats, 25 to 30 cm away.
- 02 On the tower, spray at an angle into each window so the edges are covered.
- 03 On the gallery and frame, use short bursts from several directions.
- 04 If dust or a run appears, let it dry, sand with P1200 and respray lightly.
- 05 For richer brass, spray gloss black first and gold over it.

Drying times. Typically touch-dry in 15 to 30 minutes, with the next coat after 15 to 30 minutes. Some enamel sprays must be recoated within an hour or not until 48 hours later, so check the can.

The black base and the lettering are overleaf; this step's checkpoint comes after them.

STEP 04, CONTINUED

Black base and lettering

BLACK GLOSS BASE

- 01 Gloss black shows every flaw, so give the primed base a final sand with P600 and wipe it with a barely damp cloth.
- 02 Spray a light mist coat first, then two or three light coats of gloss black, turning the base between passes.
- 03 Check it under a bright light. Sand any dust or run with P1200 once hard and respray lightly. It is clear-coated in Step 5, with the red and cream parts.

NAMEPLATE LETTERS

Once the gold is dry, brush thinned black acrylic over the letters, then wipe the raised surfaces with a damp cloth before it dries. The black stays round the letters and makes FARO easy to read.



Brush black over the letters.



Wipe the raised surface: FARO stands out in gold.

COLOUR REFERENCE



Front



Side



Rear



Three-quarter

READY TO MOVE ON WHEN

every part is evenly coloured with no bare patches or runs, the base is a deep, even black, FARO stands out in gold, and the masking tape is off.

STEP 05

Clear coat and cure

- 01 Wait at least 24 hours after the last colour coat.
- 02 Test first. Spray a little clear lacquer on a hidden inside surface, such as the inside of the cap rim, and leave it an hour. Some paints wrinkle or cloud under some clears; if it stays smooth, carry on.
- 03 Spray two light coats of clear lacquer on the red, cream and black gloss parts: band_red, cap, band_cream, tower and base.
- 04 Leave the gold uncoated; clear can dull metallic paint.
- 05 Leave everything warm, dry and dust-free for two to three days. Paint feels dry long before it is hard.
- 06 Optional: once hard, polish the lacquered parts with car polishing compound.

Drying times. Clear lacquer is typically touch-dry in about 30 minutes, with the second coat after 15 to 30 minutes. Check the can.

READY TO MOVE ON WHEN

the gloss is even and hard: a fingernail pressed on a hidden spot leaves no mark.

STEP 06

Assemble

Araldite Rapid gives about five minutes to position a joint. Mix small amounts, one joint at a time, and hold or tape each joint as it sets.

- Wait about 20 to 30 minutes before handling a glued joint or starting the next one.
- Epoxy reaches full strength the next day: leave the finished lamp overnight before lifting it by the top or testing the cap hard.
- Dry-fit each joint just before you glue it, and wipe off any squeeze-out with a cocktail stick before it sets.

THE DIFFUSERS

Each diffuser carries its number and an up arrow, raised on its inner face on a small tab below the window. The tab sits behind the tower wall, so the marks never show. Fit each one with the arrow pointing up and the smooth, curved outer face against the window. Glue them first, while the tower is still open at both ends: 1, 2 and 3 from the bottom, 4 and 5 from the top.

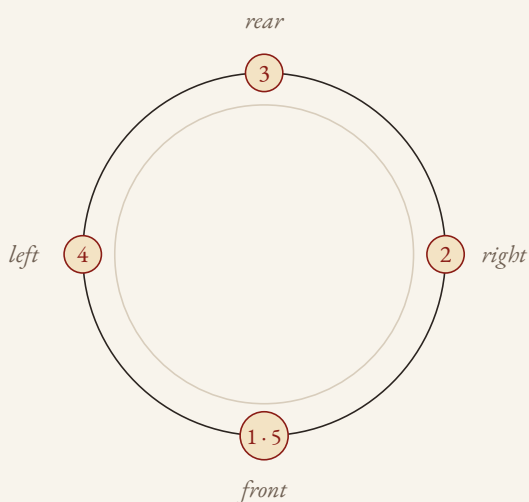


The five diffusers from inside the tower (left), and one from above (right): the curved outer face goes against the window.

STEP 06, CONTINUED

Which window is which

DIFFUSER MAP, SEEN FROM ABOVE



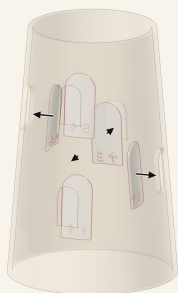
NO.	WINDOW	HEIGHT	GLUE FROM
1	Front, lowest	113 mm	Bottom
2	Right side (seen from the front)	126 mm	Bottom
3	Rear	139 mm	Bottom
4	Left side	152 mm	Top
5	Front, highest	165 mm	Top



The numbered windows, lit. Hold each diffuser to its window before gluing.

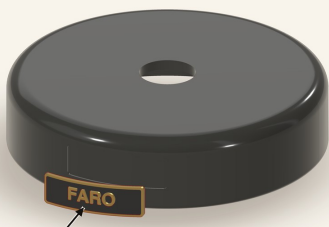
STEP 06, ASSEMBLY, 1 OF 3

Assembly order



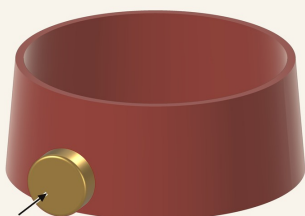
01 Diffusers into the tower

With the tower still open at both ends, glue each diffuser behind its window: arrow up, curved face against the window. Reach 1, 2 and 3 from the bottom and 4 and 5 from the top. Use tiny dabs at the edges only.



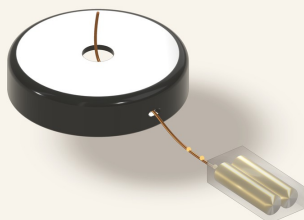
02 Nameplate

Into its recess on the front of the base.



03 Knob

Onto the front of the red band, centred. On the prototype it is decorative and doesn't turn.

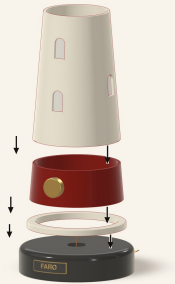


04 Thread the lights

In through the USB-C port at the back of the base and up through the hole in its top. The battery box stays outside, behind the lamp.

STEP 06, ASSEMBLY, 2 OF 3

Assembly, continued



05 Stack and glue

The cream band onto the base, then the red band, then the tower, with knob and nameplate aligned. Pull the lights up as you go, and let each joint set before the next.



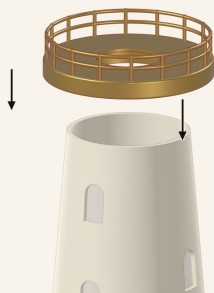
06 Coil the lights

Loosely inside the tower so some sit near each window. Hold each coil in place with a small dab of blu-tack or tape near its window. Keep them all in the tower.



07 Glow test

Before the gallery goes on, switch the lights on and check all five windows glow evenly. If one is dark or patchy, move its coil through the open top, fix it again, and test once more.

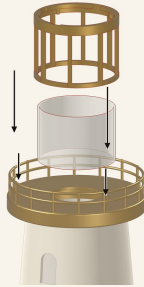


08 Gallery

Onto the tower top, centred.

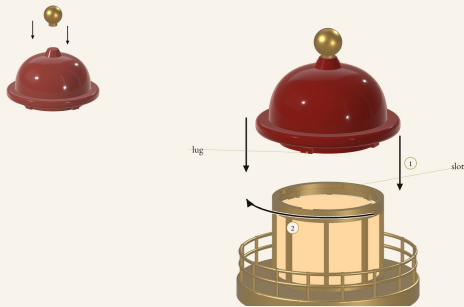
STEP 06, ASSEMBLY, 3 OF 3

Assembly, continued



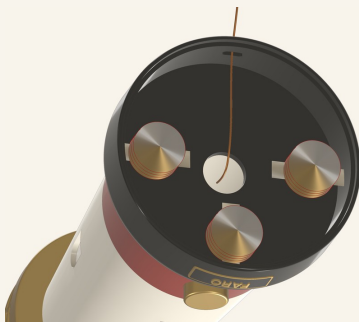
09 Lantern

Stand the empty glass on the gallery, lower the frame over it and glue the frame to the gallery. The lantern stays empty for the brightness test.



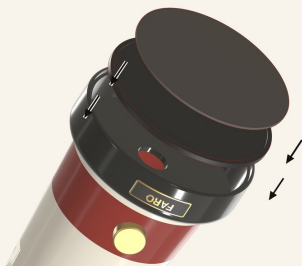
10 Finial and cap

Glue the finial into the cap. Then twist the cap on: lugs over the slots, drop it in (1) and turn clockwise to the stop (2). Do not glue it.



11 Weight

Tape a few coins low inside the base, clear of the wire.



12 Underneath

Glue the base plate into the rebate under the base, then stick on the felt, trimmed to size.

READY TO MOVE ON WHEN

the glue has set overnight, all five windows glow evenly, the cap twists on and off, and the lamp stands level on its felt.

STEP 07

Light it and photograph

The fairy lights make the windows glow, with about the light of a couple of candles. The real lamp's light would come from an LED in the lantern, so test that too.

TESTING REAL BRIGHTNESS

- 01 Twist off the cap. The opening is 49 mm across.
- 02 Drop a warm white LED puck, under 45 mm across, into the empty lantern glass. Or run a USB LED module's cable in through the USB-C port and up to a power bank.
- 03 Twist the cap back on and switch off the room lights.
- 04 Judge it, and note what you see for the lighting ODM.

WHAT GOOD LOOKS LIKE

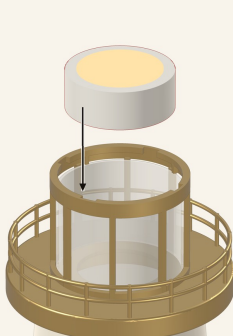
- It softly lights the nearby wall and surface, like a mood lamp.
- The frosted glass is bright without glaring.
- The light feels warm and domestic, not like a task light.

PHOTOGRAPHS

- Two sets: soft daylight with the lamp off; dusk with the lamp on.
- A plain backdrop, plus one styled shelf scene with books and a plant.
- Front, side, three-quarter, rear, lantern close-up, nameplate.
- Hide the battery box, power bank and cables out of shot.



Inside: the lights coiled in the tower, the puck in the empty lantern, coins in the base.



The puck rests on the ledge inside the gallery.



The goal: a soft, warm glow.

READY TO MOVE ON WHEN

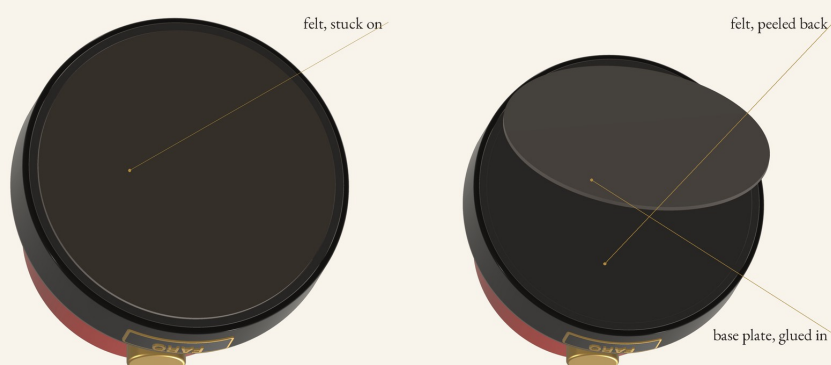
the windows and lantern glow softly, the test record is filled in, and both photo sets are taken.

REFERENCE

Troubleshooting

Almost every problem is recoverable. Let paint harden fully before sanding or recoating.

PROBLEM	FIX
Cap won't twist on, or sticks	Sand the lugs (the small tabs under the cap) and the underside of the slots lightly with P400, and test before adding more paint.
Paint run or sag	Let it harden, sand flat with P1200 and spray a light coat over it.
Rough, bumpy gloss (orange peel)	Sand gently with P1200 and polish, or add another light clear coat. Next time, lighter coats.
Dust specks	Once hard, sand the speck with P1200 and respray lightly on a still day.
Clear coat wrinkles or clouds	Let it harden, sand back and respray the colour. Next time, test the clear on a hidden surface first.
Paint clogged in the railing	Clear the gaps with a cocktail stick while still wet. Once dry, don't force the posts.
Bright LED dots in a window	Stick a small piece of baking paper behind that diffuser, or move the lights slightly away.
A window is dark or patchy	At the glow test, before the gallery goes on: move a coil closer to that window through the open top and fix it with blu-tack or tape. Later, nudge it with a long cocktail stick through the cap opening and the hole in the gallery.
A diffuser is upside down or swapped	Before the glue sets, lift it out and check its number and arrow. Once set, work it free from inside with a craft knife.
Lantern too dim, or glaring	Try a brighter or dimmer puck, or move it lower in the glass. Note the change for the ODM.
Lamp wobbles	Check the felt is flat and fully stuck, and add more coins low in the base.



The finished underside: the felt stuck on (left), and peeled back to show the base plate glued into its rebate (right).

REFERENCE

Test record

Write down what you learn before changing anything. It becomes the brief for the production lighting partner.

LED used			
Power source			
Distance to wall			
Brightness	☐ Too dim	☐ About right	☐ Too bright
Warmth	☐ Too cool	☐ About right	☐ Too warm
Glare	☐ None	☐ Mild	☐ Distracting
Diffusion	☐ Even	☐ Some hot spots	☐ Hot spots
Wall glow	☐ Weak	☐ Soft mood light	☐ Strong

NOTES FOR THE LIGHTING ODM

BEFORE YOU FINISH

- ☐ Cap still twists off
- ☐ All five windows glow
- ☐ Lantern glass empty when not testing
- ☐ Battery box and cables hidden in photos
- ☐ Changes noted for prototype two